Introducing HYN-KorPus: A Multimodal Neurocognitive Database for Korean

Jaewon Choi, Sumin Jo, Gaeun Lim, Yunju Nam*
Department of German Language and Literature, Hanyang University, Republic of Korea
Corresponding Author : Yunju Nam (yjnam05@hanyang.ac.kr)

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Introduction

- **Existing databases**
 - Cognitive neuroscience research on language processing – including corpora such as ZuCo – remains English/Indo-European-centric [1].
 - Japanese (BCCWJ) and Chinese (Beijing Sentence Corpus) – already established.
- **Prior work using database**
 - **Bensemann et al. (2022) [2]** LLM attention ↔ human eye-gaze dwell time, significant positive correlation.
 - **Barrett et al. (2015) [3]** A dependency parser trained on eye-tracking data outperformed a standard parser.
 - **Barrett et al. (2018) [4]** POS clustering possible from unlabeled eye-tracking data alone.
- **Limitation: the Korean gap**
 - Korean – only partially included in MECO Wave 1; no large-scale dedicated database exists [5].
 - Korean was not included among the 13 languages in MECO's Wave 2 expansion [6].

Korean Morphological Matters

- **Agglutinative vs. Fusional: Why morphemes matter**

Fusional language (e.g. German): Dem Student [Dative + Masculine + Singular]

VS.

Agglutinative language (e.g. Korean): 학생에게 (haksaeng-ege) [Dative]

 - Fusional language: one morpheme fuses multiple features, **inseparable**
 - Agglutinative language: each function is a **separate morpheme**, like stacking blocks

⇒ In Korean, grammatical functions live in separable morphemes, so word-level tagging cannot capture them – morpheme-level tagging is needed.

- **The Tagging Problem That Appears Prominently in Korean**
 - **"Only Korean poses an exception"** – unlike other languages, Korean adjectives correlate less with human attention, while verbs correlate more strongly [7].
 - We interpret this as a **tagging problem**: UPOS fails to reflect Korean's morphology.

(A): 아름다운 해변에 가서 (Going to the beautiful beach)

	아름다운 (Beautiful)	해변에 (Beach)	가서 (Going to)
UPOS*	아름답+ㄴ/ADJ ADJ: Adjective	해변+에/ADV ADV: Adverb	가+서/SCONJ SCONJ: Subordinating Conjunction
Sejong*	아름답/VA-I+**ㄴ/ETM** VA-I: Irregular Adjectival Predicate Stem ETM: Adnominal Ending	해변/NNG+**에/JKB** NNG: Common Noun JKB: Adverbial Case Particle	가/VV+**어서/EC** VV: Verbal Predicate Stem EC: Connective Ending

(B): 저렴한 오렌지주스 (cheap orange juice)

	저렴한 (Cheap)	오렌지주스 (Orange juice)
UPOS	저렴+하+ㄴ/VERB VERB: Verb	오렌지주스/SCONJ SCONJ: Subordinating Conjunction
Sejong	저렴/XR+하/XSA+**ㄴ/ETM** XR: Root, XSA: Adjectival Derivational Suffix ETM: Adnominal Ending	오렌지/NNG+주스/NNG NNG: Common Noun

* UPOS: universal cross-linguistic tagset
* **Sejong**: Korean morpheme-analysis standard tagset

➢ Role of Case Particles

- Tagging "해변에" as a single (ADV) hides its case, marked by particle type: –이/가 (subject), –을/를 (object), –에/에서 (adverbial)
- As its own morpheme, a case particle must be split so its grammatical function can be read separately from its lexical meaning.

➢ Predicate Conjugation

- Predicates conjugate via endings that set their function:
 e.g. 아름답- + **-ㄴ** (adnominal) → "아름다운 (beautiful)";
 가- + **-어서** (connective) → "가서 (go, and…)".
- UPOS merges stem + ending into one tag, losing the ending's function.

⇒ For these reasons, HYN-KorPus adopts **Sejong-based** morpheme annotation, rather than UPOS, as its tagging standard.

Database Status

- **Research design (2x2x2)**
 - Linguistic unit: sentence ↔ text
 - Measurement: Eye-tracking (ET) ↔ EEG
 - Modality: visual comprehension (reading) ↔ auditory comprehension (listening)
- **Current Progress**

Status	Unit	Measurement	Modality	Participants	Sentence token
Complete	Sentence	EEG	Auditory	47	420
	Text	ET	Visual	70	884
In progress	Sentence	ET	Visual	20+	210
Planned	Sentence	EEG	Visual		
		ET+EEG	Visual		
	Text	EEG	Auditory		
		ET+EEG	Visual		

- **Sentence stimulus design**
 - Grammatical/ungrammatical pairs from a grammaticality-judgement corpus.
 - Error types designed across four levels: morphological, morphosyntactic, syntactic, semantic.

Example: Morphosyntactic Violation (를 → 가)

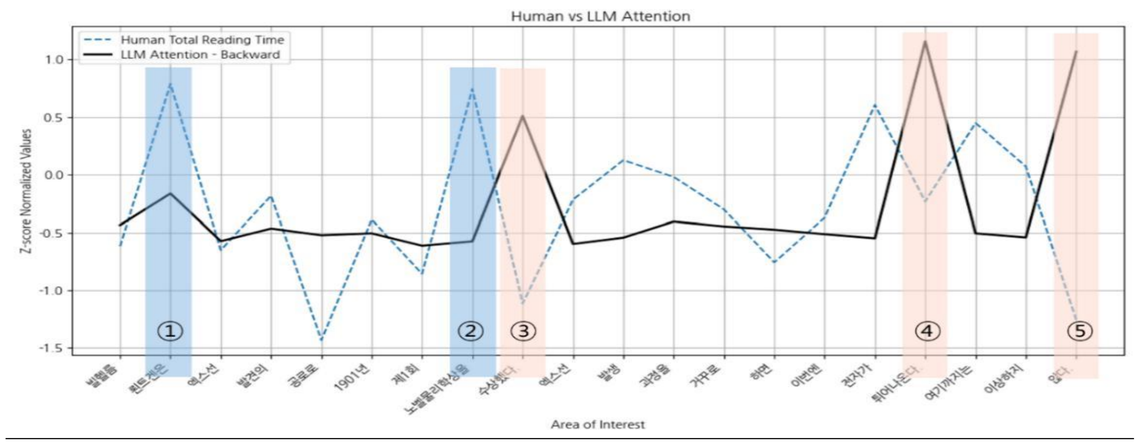
(Grammatical)	**나는 그 영화를 보았다.** Naneun geu yeonghwa-**reul** boatda. I watched that movie	
(Ungrammatical)	***나는 그 영화가 보았다.** *Naneun geu yeonghwa-**ga** boatda.	"가" is fine with intransitives (e.g. 재미있다; "interesting"), but clashes with a transitive verb like 보다 ("see"), which requires an object.

 - By contrasting a morphosyntactically grammatical sentence with its ungrammatical counterpart at the particle **(를 vs. 가)**, this pair is designed to capture **ERP/P600** signals – allowing us to examine what role case particles play in real-time Korean sentence comprehension.
- **Text stimulus design**
 - Extends the sentence-level design to the text level, with **9 passages** spanning three genres: **non-fiction, novel openings, and complete short stories**.
 - The same 9 passages will also be narrated as audiobooks for the planned **Text·EEG·Auditory** condition, enabling comparison of the same text **across modality and measurement**.
- **Data release**
 - Participants recorded to data: **140+**
 - Data will be released incrementally on **OSF** as preprocessing is completed.

Pilot Analysis of HYN-KorPus (Eye-Tracking Data)

RQ: Does LLM attention in Korean reading show patterns similar to human eye-movement processing (reading time·regression) [8]?

- ***Attention: the relative weight a transformer assigns to each token in a sentence – used here as a proxy for how much attention a human pays to that word.***



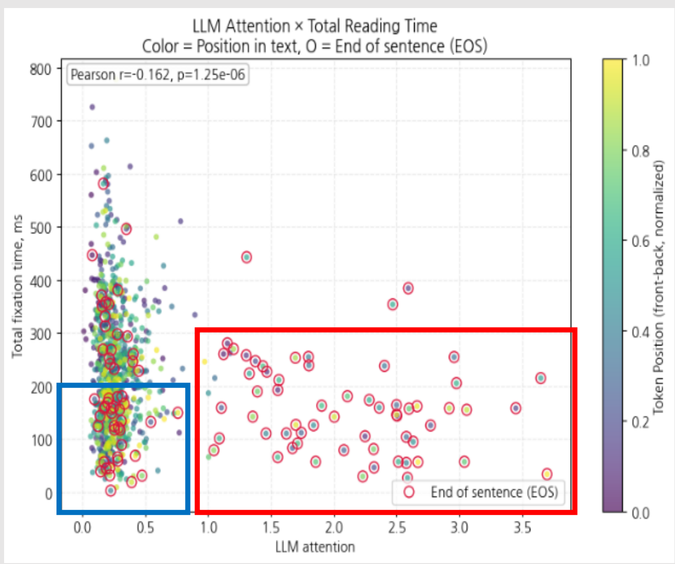
	①	②	③	④	⑤
Korean	룐트겐은	노벨물리학상을	수상했다	튀어나온다	않다
Romanization	Roentgen-eun	Nobel-mulli-haksang-eul	Susang-haet-da	Twieona-onda	An-ta
Gloss	Roentgen(unfamiliar proper noun) + topic marker	"Nobel Prize in Physics" + object marker	"received (the prize)" (predicate)	"protrudes" (predicate)	"not" (negative predicate)

➢ Figure 1. Human reading time vs. LLM attention at 5 points in a sentence

- ①, ② (blue shading): Points where human reading time is elevated – due to low-frequency vocabulary, proper nouns.
- ③, ④, ⑤ (red shading): Sentence-final predicates – unlike the English wrap-up effect (reading time increases as meaning is reconstructed at sentence end), Korean shows the opposite: reading time **decreases** at the predicate → a reversal of the wrap-up effect [9].
- Japanese, another head-final language, likewise shows no clause wrap-up effect [10].

➢ Figure 2. LLM attention x total fixation time across the full passage

- Blue box (attention↓, RT↓): similar to human eye-movement processing
- Red box (attention↑, RT↓): opposite of human eye-movement processing: LLM attends, but humans move past quickly
- Suggests Korean readers may follow a "resolution-based" rather than an "accumulative" sentence-processing strategy, indicating that the wrap-up effect is not universal but varies by language type



Conclusion & Discussion

- Morpheme-level annotation (*Sejong-based*) revealed a Korean-specific processing pattern that UPOS would have missed – a reversal of the attention-based wrap-up effect.
- Future work will extend to Korean L2 learners to compare processing mechanisms between native and non-native speakers, helping identify which patterns are language-universal versus Korean-specific – further broadening the range of analyses HYN-KorPus can support, from attention-based to surprisal-based and other computational approaches.
- Upon completion of preprocessing, the corpus will be released freely on OSF – usable as a benchmark for psycholinguistics, computational linguistics, language education, and Korean LLMs.

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